

0 5 . 1

Place a tick (✓) in the box next to the sequence that shows the correct order of magnitude of these measurements.

[1 mark]

50 nm < 0.5 μ m < 5×10^{-2} mm < 0.5×10^{-5} m ☐

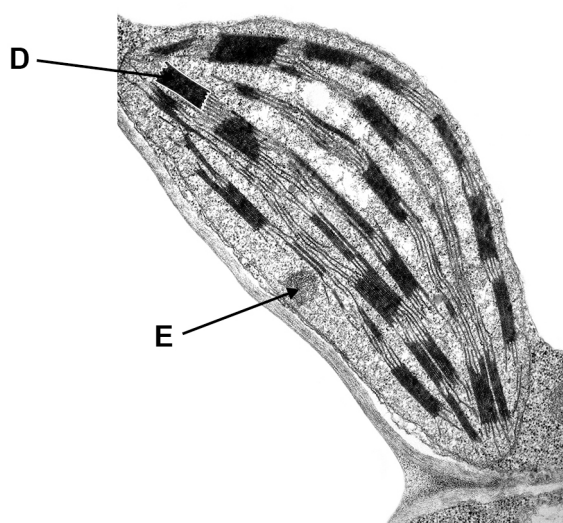
50 nm < 0.5 μ m < 0.5×10^{-5} m < 5×10^{-2} mm ☐

0.5 μ m < 50 nm < 0.5×10^{-5} m < 5×10^{-2} mm ☐

0.5 μ m < 50 nm < 5×10^{-2} mm < 0.5×10^{-5} m ☐

Figure 3 is an electron micrograph of a chloroplast.

Figure 3



0 5 . 2

Identify structures labelled **D** and **E**.

[2 marks]

D _____

E _____



0 5 . 3 The detail shown in **Figure 3** would **not** be seen using an optical microscope.

Explain why.

[2 marks]

0 5 . 4 Name an organelle found in both a chloroplast and a prokaryotic cell.

[1 mark]

0 5 . 5 A scientist determined the volume of a plant cell and the volume of organelles it contained.

They found:

- the volume of a plant cell is $17\,500\ \mu\text{m}^3$
- the volume of all the mitochondria in a plant cell is $262.5\ \mu\text{m}^3$
- the volume of all the mitochondria and all the chloroplasts in a plant cell is 44.1% of the volume of a plant cell.

Use this information to calculate the volume of all the chloroplasts in a plant cell.

[2 marks]

Answer = _____ μm^3

Question 5 continues on the next page

Turn over ►



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A biologist separated cell components to investigate organelle activity. She prepared a suspension of the organelles in a solution that prevented damage to the organelles.

Describe **three** properties of this solution and explain how each property prevented damage to the organelles.

[3 marks]

Property 1 _____

Explanation

Property 2 _____

Explanation

Property 3 _____

Explanation

11

